

# Federated Learning: Securing AI Models Using Decentralised Data

Machine learning models must be developed with care. When based on centralised data, they are prone to a single point of failure and can attract cyberattacks leading to data breaches. Federated learning, which can be used to build models based on decentralised data, addresses this challenge successfully.



**A**t its core, artificial intelligence (AI) encompasses algorithms designed to interpret, predict, and potentially alter data streams to facilitate decision-making processes that traditionally required human intervention. These complex algorithmic structures are often built upon machine learning models that train on large datasets, enabling the AI to learn from patterns and improve its accuracy over time.

AI is transforming numerous sectors, enabling enhanced efficiency, improved accuracy, and innovative

solutions to longstanding challenges. Here are some of its practical real-world applications.

## Healthcare

- *Patient diagnostics:* AI helps to interpret medical imaging faster and often more accurately than human radiologists, aiding in the early detection of conditions like cancer.
- *Robotic surgeries:* Sophisticated robots assist surgeons in performing delicate operations with higher precision and control,

potentially reducing recovery times.

- *Personalised medicine:* AI analyses individual health profiles to tailor drug regimens and therapies that optimise patient outcomes.

## Finance

- *Automated trading systems:* AI helps analyse market data to execute trades at optimal times, aiming to maximise returns or minimise risks.
- *Fraud prevention:* By learning from transaction data, AI can identify patterns indicative of fraudulent activities, securing users against